(a)

Rainfall: varying steepness

Scalar

1.0
0.8
0.6
0.4
0.2
0.0

s_{Rain}=0.2 (0.2x)
s_{Rain}=0.5 (0.5x)
s_{Rain}=1.0 (1xdefault)
s_{Rain}=1.5 (1.5x)
s_{Rain}=2.0 (2x)

0 5 10 15 20

Rainfall ($mm\ day^{-1}$)

(b)

Rainfall: varying midpoint

Scalar

1.0
0.8
0.6
0.4
0.2
0.0

$Rain_0$=0.6 (0.2x)
$Rain_0$=1.5 (0.5x)
$Rain_0$=3.0 (1xdefault)
$Rain_0$=4.5 (1.5x)
$Rain_0$=6.0 (2x)

0 5 10 15 20

Rainfall ($mm\ day^{-1}$)

(c)

Soil moisture: varying steepness

s_θ=2 (0.2x)
s_θ=5 (0.5x)
s_θ=10 (1xdefault)
s_θ=15 (1.5x)
s_θ=20 (2x)

Scalar

1.0
0.8
0.6
0.4
0.2
0.0

0.0 0.1 0.2 0.3 0.4 0.5 0.6

Volumetric soil water ($m^3\ m^{-3}$)

(d)

Soil moisture: varying midpoint

θ_0=0.067 (0.2x)
θ_0=0.168 (0.5x)
θ_0=0.335 (1xdefault)
θ_0=0.503 (1.5x)
θ_0=0.670 (2x)

Scalar

1.0
0.8
0.6
0.4
0.2
0.0

0.0 0.1 0.2 0.3 0.4 0.5 0.6

Volumetric soil water ($m^3\ m^{-3}$)